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MN50755 Data Mining and Machine Learning

Prepared for the Chief Financial Officer

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Executive Summary

Three years of approved mortgage data reveal two predictable problems, not knowing which deals will fall through and not knowing when the rest will be funded. Using a calibrated Random Forest for completion probability and Gradient Boosting for timing, the analysis shows that completing mortgages can be ranked correctly above 77.3% of the time. The analyst's original model managed 56.8%, barely better than a coin toss.

Gradient Boosting reduces average timing error to 10.73 days, down from 13.52 days in the original approach. Applied to the 3,885 applications still open on 1 September 2024, a pipeline worth £649.6 million, a bootstrap simulation puts expected advances for the following two months at £80.7M with a 90% confidence range of £76.0M to £85.5M. If completion rates fall ten percentage points below forecast, that figure drops to £69.8M. The gap between those two numbers, £10.9M, is the liquidity buffer this analysis recommends.

1. Data Exploration and Preparation

The dataset covers approved applications from Q3 2021 to Q3 2024, spanning loan size, property valuation, location, business channel, applicant demographics, property characteristics and advance date where recorded. Before any model could be trained, four problems in the raw data had to be resolved. Left unaddressed, each one would have quietly distorted every result that followed.

Right-Censoring

Across all twelve completed quarters, the completion rate sits in a narrow band of 70% to 76%, with a historical mean of 73.1%. Then Q3 2024 registers 27.6%, a figure that looks alarming but is nothing of the sort. Applications approved in June, July and August 2024 simply had not had time to complete by the date the data was extracted. Labelling them as non-completions would teach every subsequent model that recent approvals are inherently more likely to fail. All Q3 2024 observations were therefore stripped from the training data.

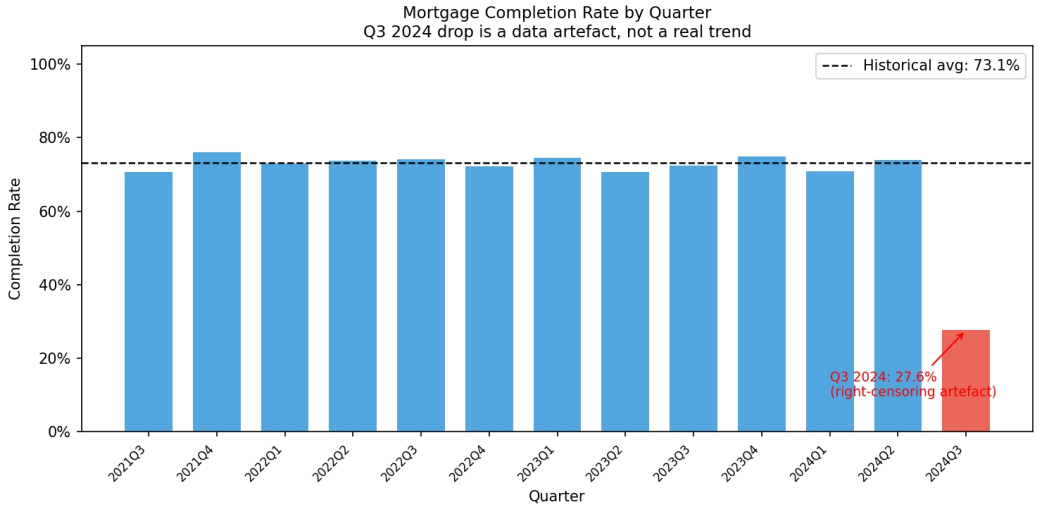


Figure 1. Completion rate by quarter. The Q3 2024 reading is a right-censoring artefact and was excluded from all model training.

Property Age Signal

Completed mortgages sit on substantially older properties than non-completed ones, a median of roughly 80 years versus 45 years. That gap turns out to be the strongest signal in the entire dataset. About 10% of records have no property age recorded, and that missingness is not random; whether an age is missing is itself a statistically meaningful predictor. Simply substituting zero would corrupt this signal entirely. A binary flag was added to mark the absence, allowing the models to learn a separate completion pattern for that group.

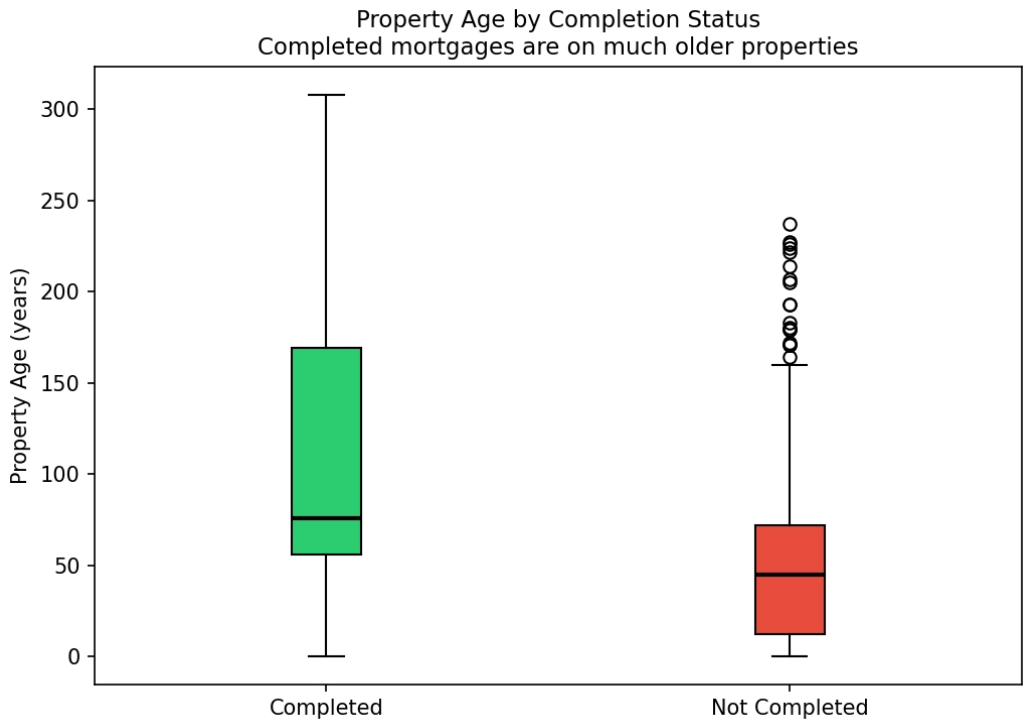


Figure 2. Property age by completion status. The gap between completed and non-completed cases is the largest observed across any variable.

LTV and Completion

Loan-to-value ratio was engineered as a single credit metric rather than entering loan and valuation as separate inputs. The data surface a puzzling pattern; completion rates rise with LTV, from 65% for borrowers below 50% LTV to 79% above 90%. The most plausible explanation is borrower selection. To get approved at high LTV, applicants must clear a more demanding affordability hurdle; once they do, they are committed. Low-LTV borrowers have more optionality and can walk away or delay with less cost. This interpretation is tentative but consistent across all subsequent model outputs.

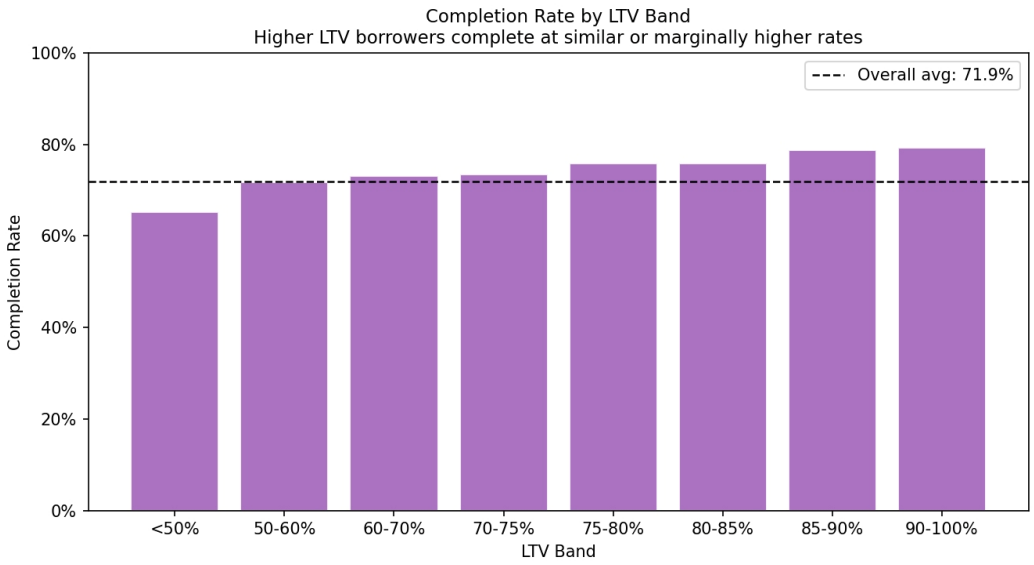


Figure 3. Completion rate by LTV band. The monotonically positive relationship is consistent with a borrower selection effect.

Completion Time Distribution

Completion times are right-skewed, mean 28.1 days, median 27, with a tail stretching past 80 days. Fitting a standard linear regression directly to these figures violates the model's assumptions and inflates errors at both ends. A log transformation was applied before modelling and predictions were converted back to days using a smearing correction (Duan, 1983).

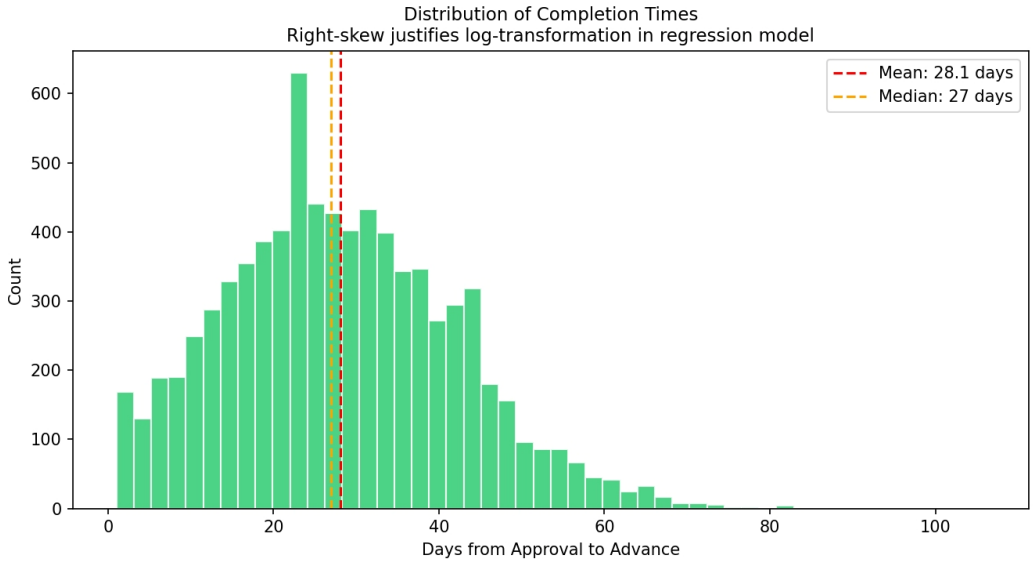


Figure 4. Distribution of completion times. Right-skew justifies log-transformation of the outcome in both regression models.

2. Flaws in the Baseline Modelling Approach

The analyst's logistic model predicts completion using three variables: country, business source and raw loan amount. Property age and LTV, the two strongest predictors by a wide margin, are absent. The consequence is an AUC of 0.568. An AUC of 0.5 means random guessing; at 0.568 the model is barely distinguishing anything and could not reliably support portfolio management or cashflow forecasting.

The missing value treatment makes things worse. Replacing all missing values with zero works only if zero is meaningless in context. Here it is not. Zero is a real category code for country (England), property type (flat) and area (urban), so filling gaps with zero tells the model that every unknown observation is English, flat-dwelling and urban, a false assumption affecting roughly 10% of the dataset across four variables. For property age, zero implies a newly built property, which distorts the dominant predictive signal. Neither baseline model is validated on a hold-out sample, so their published fit statistics are optimistic by construction and provide no evidence of generalization.

The timing model contains two separate problems, one at the code level and one at the design level.

The first is a coding error. The variable y is set to the completion indicator for model 1 and is never reassigned before model 2 is fitted. The OLS call therefore predicts a Boolean completion flag rather than days to advance. The intended timing model does not exist in the analyst's code at all. To produce a meaningful comparison, the analyst's timing baseline was reconstructed correctly in this analysis, restricting it to completed cases with the three original features and no log transformation.

The second problem is a design error that would persist even with the coding fix. Including non-completing cases in a timing model and assigning them a time_to_complete of zero via fillna(0) produces labels that are indistinguishable from a mortgage that funded on the same day it was approved. The model conflates the absence of completion with a near-instantaneous one, corrupting roughly 30% of training labels. Timing models must be estimated on completed cases only, since they answer a conditional question, given that a mortgage advances, how long does it take? The corrected analyst baseline, fitted on completed cases with the original three features and no transformation, produces an RMSE of 13.52 days.

3. Improved Linear Probability Model

The improved logistic regression fixes each of the analyst's deficiencies. Predictors include LTV, combined household income, property age with its missingness indicator, applicant age, approval month, and one-hot encoded versions of country, property type, business source and area. Right-censored Q3 2024 observations were removed. The model was trained at a random 80% of the cleaned data and evaluated on the remaining 20% it had never seen.

What the Coefficients Say

Property age carries a standardized coefficient of approximately 1.75, more than seven times larger than LTV (0.25) or combined income (0.20). Both of those increase completion probability, consistent with the data exploration findings. Several property types and business source dummies carry small negative coefficients. One detail worth flagging is the Property_Age_missing indicator is itself negative, meaning applications with no recorded property age complete at lower rates than the baseline. That points to a gap in data collection that warrants operational investigation.

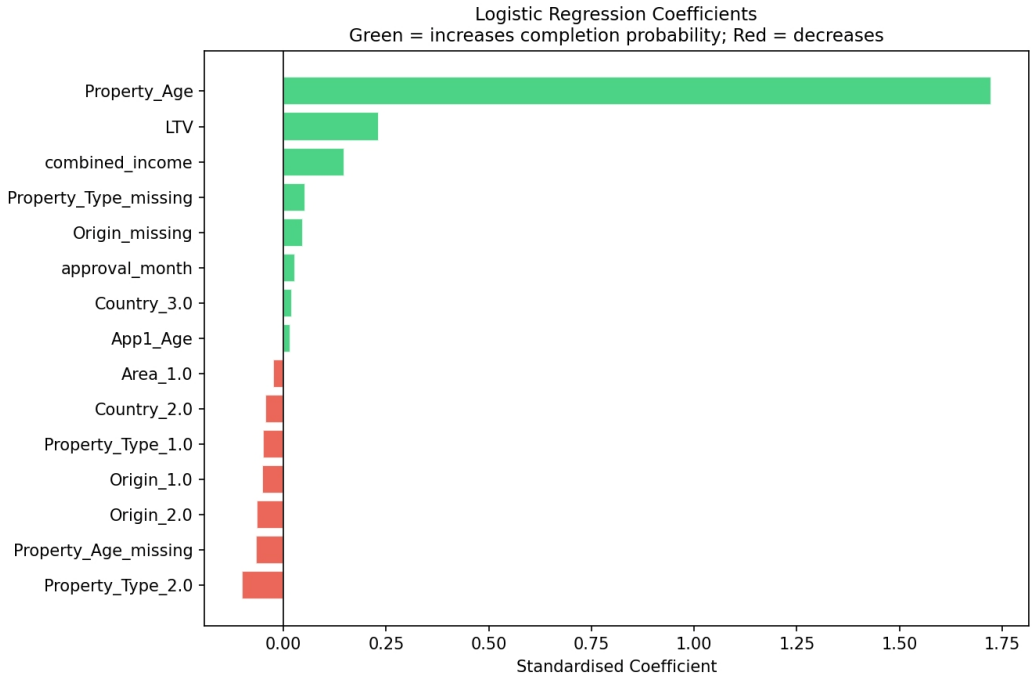


Figure 5. Standardized logistic regression coefficients. Green bars increase completion probability; red bars decrease it.

Performance Against the Baseline

The improved model achieves AUC = 0.773 on the test set, compared with 0.568 for the analyst's baseline. The model correctly ranks a completing mortgage above a non-completing one 77.3% of the time, a meaningful and usable signal for prioritization and cashflow planning. Five-fold cross-validation returned to a mean AUC of 0.777 with a standard deviation of 0.008, confirming the result is stable.

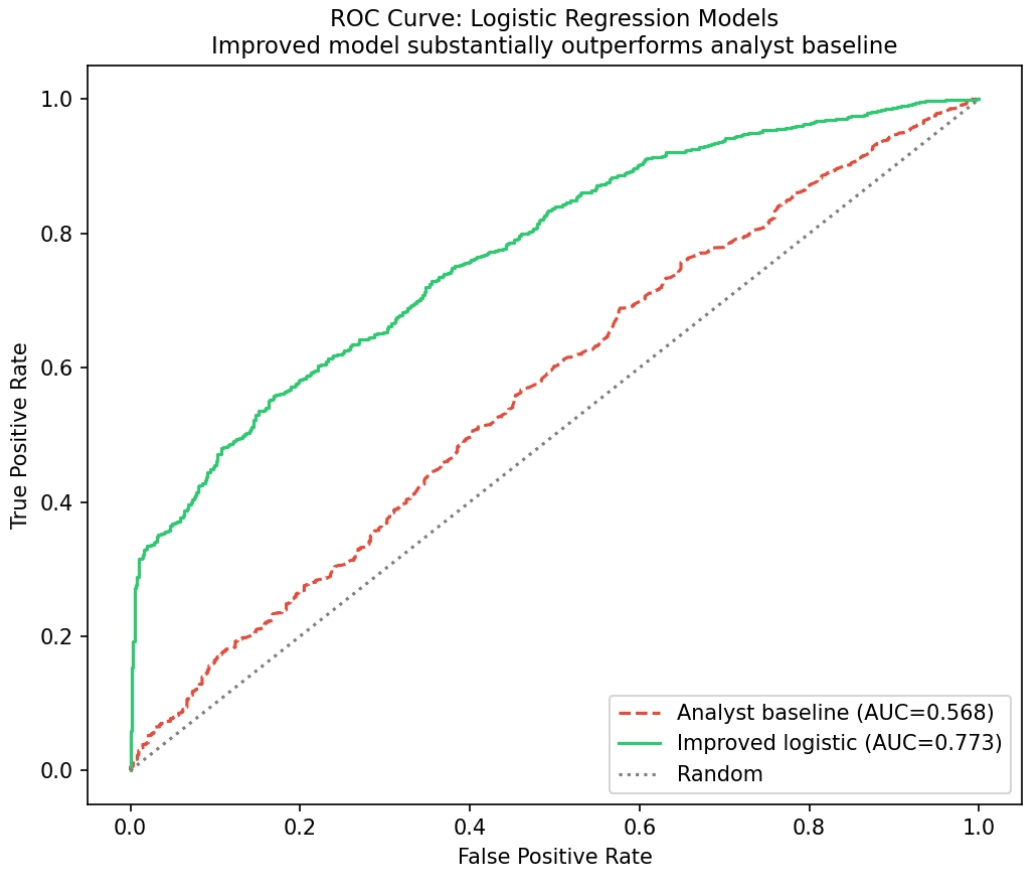


Figure 6. ROC curves for the analyst baseline and improved logistic regression. The improved model (AUC = 0.773) substantially outperforms the baseline (AUC = 0.568).

4. Improved Linear Timing Model

For timing, the sample is restricted to completed cases only, 7,851 observations after cleaning. The outcome is log-transformed before modelling and back-transformed using a Duan (1983) smearing correction. Predictors mirror those in Section 3, with joint applicant status added since dual-income households may be better placed to manage delays.

Test RMSE is 10.91 days and R^2 is 0.376, an improvement on the analyst's 13.52 days. The R^2 of 0.376 is sometimes misread as disappointing. It should not be. The 62% variance the model does not explain reflects solicitor availability, survey outcomes and property chain complexity, things that are invisible at the application stage. The model captures what is knowable; the rest is genuine noise.

The residual plot shows mild compression at the extremes, an expected consequence of log-retransformation rather than evidence of misspecification. Very short completions are nudged upward slightly and very long ones nudged down, both pulled toward the mean on the log scale.

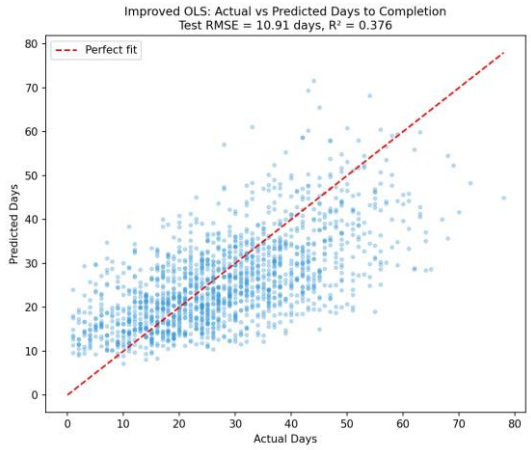


Figure 7. Actual vs predicted completion days. RMSE = 10.91, $R^2 = 0.376$.

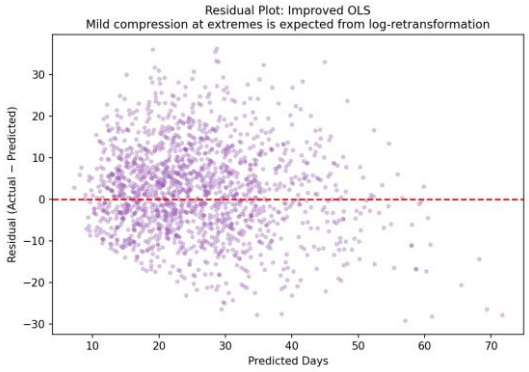


Figure 8. Residual plot. Mild compression at extremes expected from log-retransformation; no systematic directional bias.

5. Non-Linear Probability Model: Random Forest

Random Forest builds 500 decision trees, each trained on a different bootstrap sample and constrained to consider a random subset of variables at each split. No tree sees the full picture; averaging across all of them iron out the idiosyncrasies any single tree would overfit. Non-linear relationships and feature interactions are captured without needing to be specified in advance. Hyperparameters were selected by five-fold cross-validation.

Feature Importance

Property age accounts for 72% of total importance, the same signal that dominated the logistic coefficients, now confirmed by a completely different modelling framework. LTV contributes 8%, combined income 6%, applicant age 4%. Everything else is marginal. The concentration in a single variable is striking enough to warrant caution; part of the property age signal may reflect the pattern that new-build properties, which tend to have missing age values, complete more reliably, rather than a true age effect. The two stories are difficult to separate without additional data.

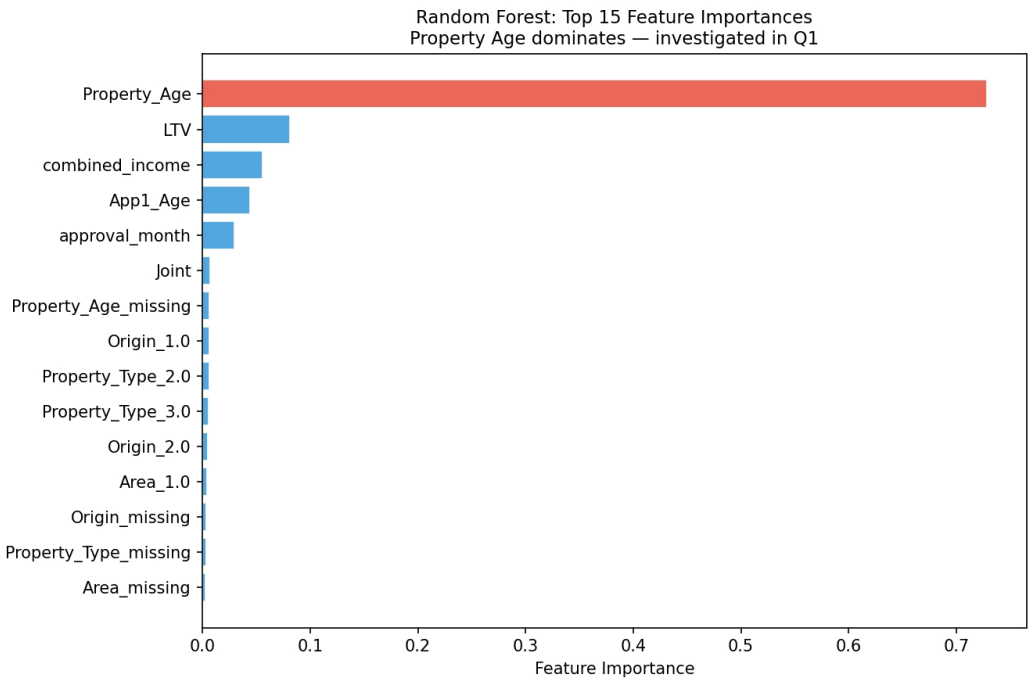


Figure 9. Random Forest feature importances. Property age dominates at 72%, consistent with the logistic regression coefficients.

Comparison with the Logistic Model

The Random Forest achieves AUC = 0.772. The improved logistic model reached 0.773. That difference is negligible, with cross-validation noise, and the conclusion it supports is worth stating plainly. A flexible non-linear model adds nothing that the simpler linear specification does not already capture. For a regulated lender, where the FCA expects credit decisions to be explainable at the individual level, this convergence is an argument for preferring the logistic model in live deployment; it is auditable, interpretable and performs identically.

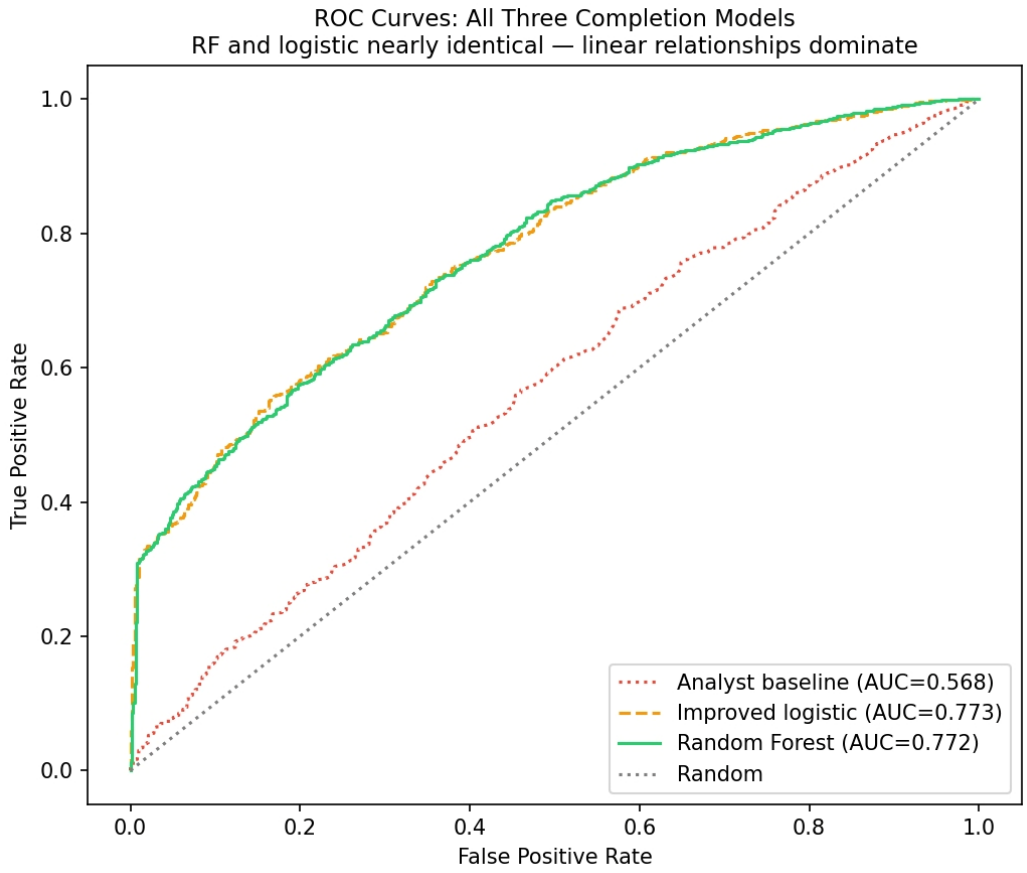


Figure 10. ROC curves for all three completion models. The Random Forest and improved logistics are near-identical, indicating that linear relationships dominate this dataset.

Calibration

Discrimination and calibration measure different things. A model can rank completing cases reliably while being systematically wrong about the actual probabilities it assigns. The uncalibrated Random Forest overestimates completion probability in the mid-range by around 10 to 15 percentage points, which would inflate the Section 7 cashflow estimate correspondingly. Isotonic regression calibration (Zadrozny and Elkan, 2002) corrects this within a cross-validated framework, without meaningfully affecting AUC (0.7723 to 0.7721). A minor non-monotone dip visible around p = 0.45 is a known artefact of isotonic fitting on small probability bins and affects fewer than 3% of open-pipeline cases. Calibrated probabilities are used throughout Section 7.

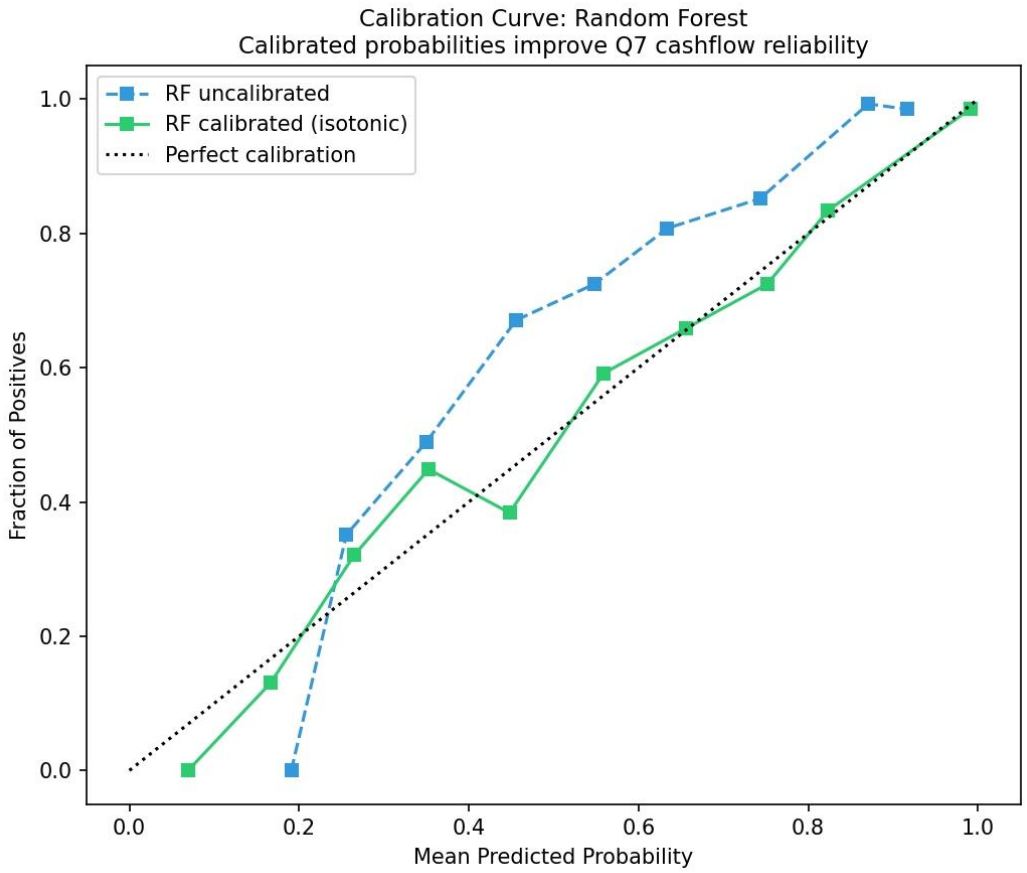


Figure 11. Calibration curve for the Random Forest. Isotonic correction brings predicted probabilities close to the diagonal, essential for reliable use in the cashflow simulation.

Partial Dependence: LTV

The partial dependence plot shows how completion probability changes as LTV varies, with all other variables held at their observed values. The relationship runs from 0.49 at LTV 0.40 to 0.59 at LTV 0.82, positive and monotonic throughout, consistent with the selection effect hypothesis and the raw data analysis in Section 1.

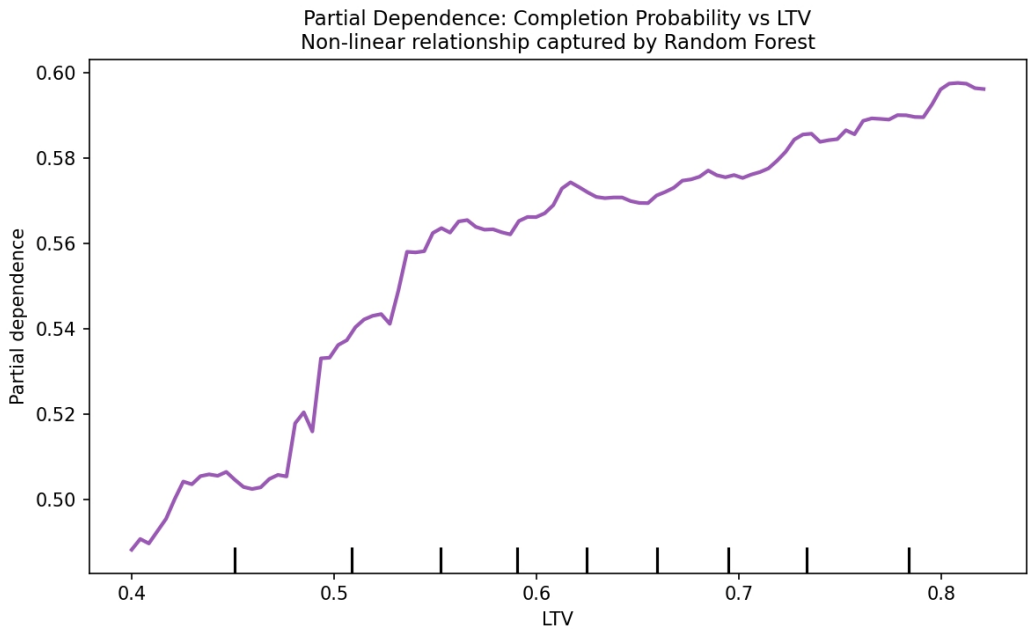


Figure 12. Partial dependence of completion probability on LTV. The monotonically positive relationship is consistent across all models.

6. Non-Linear Timing Model: Gradient Boosting

Gradient Boosting builds trees sequentially rather than independently. Each new tree is fitted to the errors left by the previous ensemble, allowing the model to reduce bias as well as variance. A learning rate of 0.05 limits how much any single tree can move the result, functioning as a brake against overfitting. Tree depth was capped at four levels and 300 estimators were used, all settings chosen by five-fold cross-validation. Training was restricted to completed cases with a log-transformed outcome, matching the OLS setup in Section 4.

The LTV–Age Reversal

For timing, LTV leads to over 55% importance; property age drops to 17%. In the completion probability model, that order was reversed. The divergence suggests the two problems genuinely have different primary drivers. Property age shapes whether a transaction completes; borrower leverage shapes how quickly it funds once it is on track. Whatever the mechanism, this finding validates the decision to model the two questions separately rather than combining them.

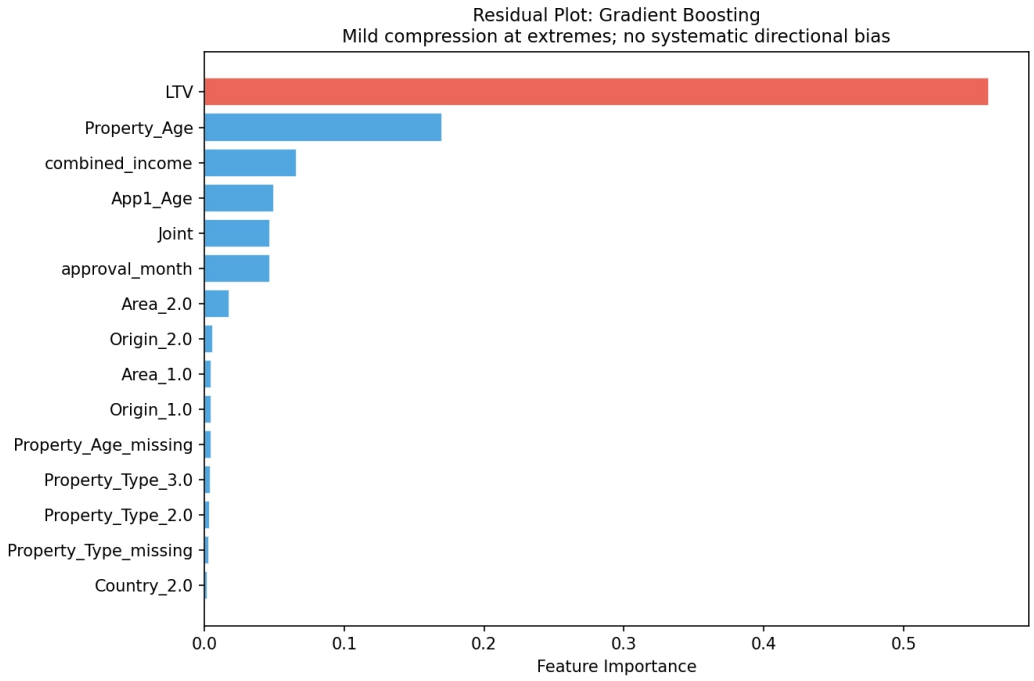


Figure 13. Gradient Boosting feature importances for timing. LTV dominates at 55%, contrasting sharply with property age's 72% share in the completion model.

Performance

Gradient Boosting achieves a test RMSE of 10.73 days and R^2 of 39.7%, improving on the improved OLS (10.91 days, 37.6%) and the analyst baseline (13.52 days). The improvement is modest but consistent across cross-validation folds, confirming it reflects genuine signal rather than a favorable split. Residuals are centered on zero with no systematic directional bias.

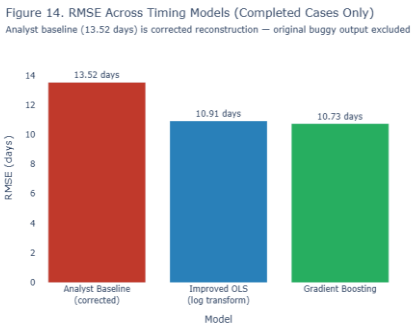


Figure 14. RMSE comparison across all three timing models.

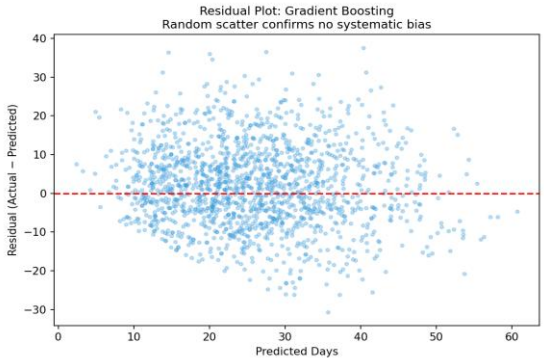


Figure 15. Gradient Boosting residuals. Mild compression at extremes; no systematic directional bias.

7. Bootstrap Cashflow Simulation from September–November 2024

At the snapshot date of 1 September 2024, 3,885 approved applications remained open, representing a total potential loan book of £649.6 million. The question for treasury is not whether those deals will eventually fund, but how much is likely to advance in the next two months and how wide the uncertainty genuinely is.

Open Pipeline Completion Probabilities

Figure 16 shows the calibrated completion probabilities across the open pipeline. The distribution is bimodal: a lower cluster between 0.20 and 0.40 for higher-risk cases, and a larger concentration between 0.65 and 0.80. The mean is 0.569, materially below the historical 71.9% completion rate. Many of these applications were approved recently and will not advance within a two-month window even if they are eventually complete. Treating the open pipeline as if it behaved like the historical dataset would overstate near-term cashflows significantly.

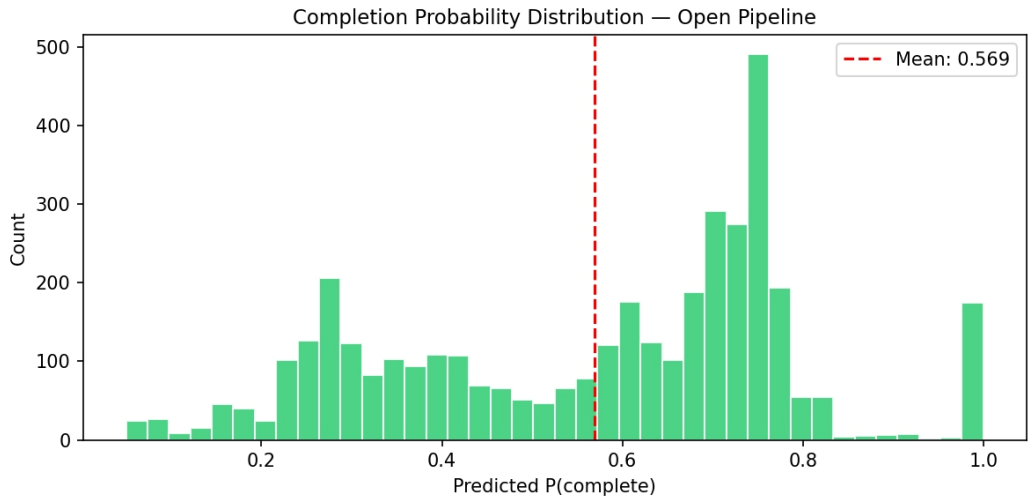


Figure 16. Calibrated completion probability distribution across 3,885 open-pipeline applications. Mean P(complete) = 0.569.

Bootstrap Methodology

A bootstrap simulation with 2,000 replications was run. In each iteration, every open application was independently simulated as completing or not using a Bernoulli draw from its calibrated probability. For applications that are completed in the simulation, the Gradient Boosting model predicted days to advance. Gaussian noise was then added on the log scale, with standard deviation equal to the model's out-of-sample RMSE, to reflect that individual timing predictions carry uncertainty beyond the point estimate. Adding noise on the log scale rather than directly to days preserves the right-skewed structure of timing errors and avoids physically impossible negative values for fast-completing cases. Advance dates were computed and loan amounts for applications advancing before 1 November 2024 were summed. Repeating this 2,000 times yields a distribution of plausible aggregate cashflows (Efron and Hastie, 2016).

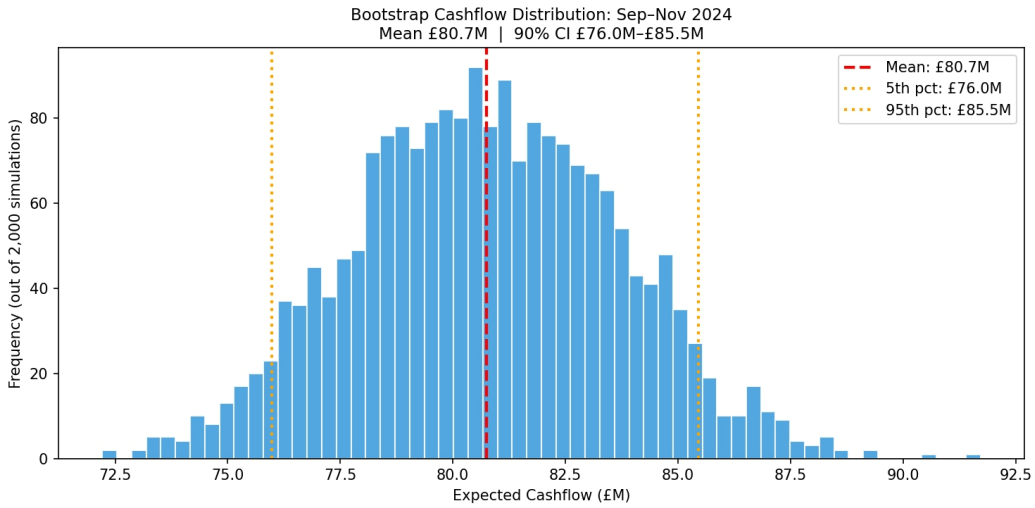


Figure 17. Bootstrap cashflow distribution, September–November 2024. Mean £80.7M, 90% CI £76.0M–£85.5M across 2,000 simulations.

Results and Liquidity Recommendation

The simulation yields a mean expected cash flow of £80.7M, standard deviation £2.93M, with a 90% confidence interval of £76.0M to £85.5M, and a planning range of £9.5M. The near-symmetrical shape reflects the aggregation of thousands of independent Bernoulli draws, consistent with central limit theorem behavior at portfolio scale. Individual predictions are uncertain; in aggregate that uncertainty largely cancels.

Three scenarios were modelled. Under the base case, £80.7M is the best estimate. If completion rates run ten percentage points below prediction, a plausible outcome under a conveyancing slowdown or an adverse rate movement, expected advances fall to £69.8M. An upside of five percentage points produces £85.0M. The stress gap is £10.9M, and that is the reserve treasury should plan around to remain solvent if conditions deteriorate without warning.

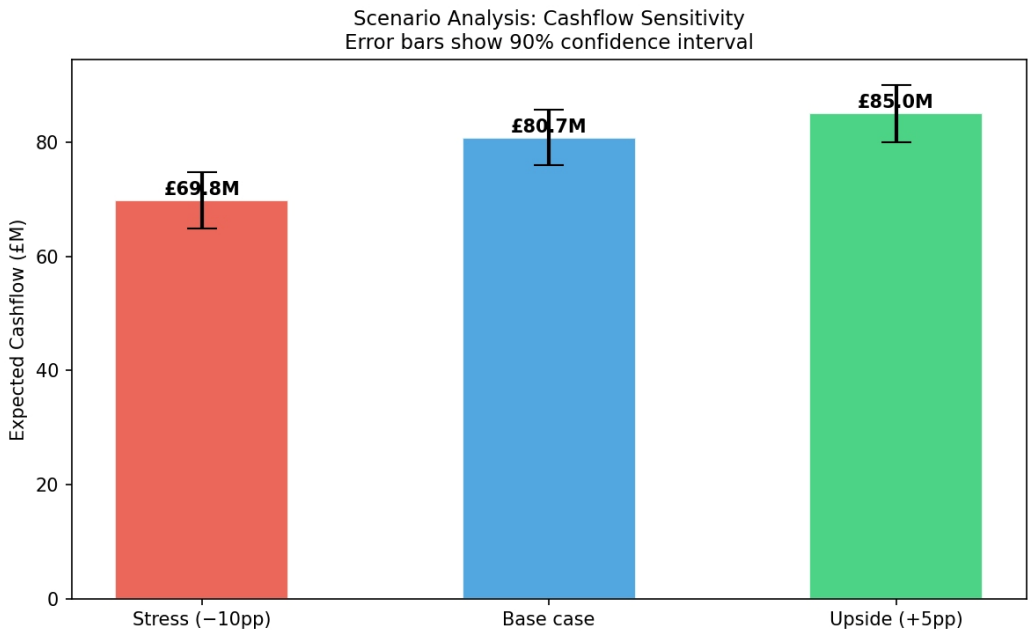


Figure 18. Scenario analysis. A stress scenario of minus 10pp reduces expected cashflow by £10.9M relative to the base case, defining the minimum recommended liquidity buffer.

A point estimate of £80.7M would not support this kind of planning. The bootstrap produces a range, a stress case and an upside. Each requires a different treasury response. That is the practical value of the method.

8. Limitations

The models were trained on data up to June 2024. If market conditions have shifted since then, rising rates or a cooling housing market, the historical patterns underpinning the predictions may no longer hold. The models cannot detect that shift and will continue forecasting based on what they learned from earlier periods.

No macroeconomic variables are included. Interest rate levels, house price trends and mortgage market sentiment all plausibly affect completion rates, but none appears in the application data. A Bank Rate increase during the forecast window could reduce completion intent across the portfolio simultaneously, a correlated shock that the bootstrap's independence assumption would understate. The £9.5M confidence interval should be read as a floor on uncertainty under normal conditions, not a ceiling under stress.

The non-random missingness in property age is handled via an indicator variable but its root cause is unresolved. If the proportion of applications with missing property age changes materially, the model's ability to learn from that indicator may degrade over time.

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Appendix

A. Dataset Summary Statistics

The table below summarises the key numeric variables in the cleaned training dataset (11,207 observations after removing right-censored Q3 2024 records and negative completion times). Completion time statistics cover completed cases only (8,134 observations).

Variable	Mean	Median	Std Dev	Min	Max
Loan amount (£)	181,395	141,000	142,309	11,000	2,216,000
Property valuation (£)	292,350	214,000	215,111	17,700	2,909,600
LTV ratio	0.60	0.60	0.13	0.20	1.00
Applicant 1 age (yrs)	45.2	45	14.1	18	75
Applicant 1 income (£)	23,998	20,000	17,788	1,432	340,440
Property age (yrs)	84.5	72	66.3	0	308
Completion time (days)	27.7	27	14.1	0	106

Table A1. Descriptive statistics for key numeric variables. Training data only (Q3 2024 excluded).

B. Completion Rate by Country

Completion rates are broadly consistent across the four nations, ranging from 71% to 76%. Northern Ireland carries the smallest volume but the highest average loan size, reflecting higher-value property transactions relative to portfolio share.

Country	Applications	Completion Rate	Avg Loan (£)
England	8,818	72.1%	£181,601
Northern Ireland	393	73.0%	£179,018
Scotland	601	69.1%	£187,547
Wales	300	66.7%	£167,513

Table A2. Completion rate and average loan size by country.

C. Model Performance Summary

The table below consolidates performance metrics across all models tested. AUC is reported for completion probability models; RMSE (days) for timing models. All metrics are evaluated on held-out test sets unseen during training.

Model	Task	Metric	Score	vs Analyst Baseline
Analyst baseline (corrected)	Completion probability	AUC	0.568	—
Improved Logistic Regression	Completion probability	AUC	0.773	+0.205
Random Forest (calibrated)	Completion probability	AUC	0.772	+0.204
Analyst baseline (corrected)	Timing (days)	RMSE	13.52	—
Improved OLS (log transform)	Timing (days)	RMSE	10.91	−2.61
Gradient Boosting	Timing (days)	RMSE	10.73	−2.79

Table A3. Consolidated model performance. All metrics on held-out test sets. Analyst baseline corrected for the unreassigned y variable bug.

D. Bootstrap Simulation Parameters

The following parameters were used in the Section 7 bootstrap cashflow simulation.

Parameter	Value
Number of iterations	2,000
Snapshot date	1 September 2024
Forecast window	1 September – 1 November 2024
Open pipeline applications	3,885
Total pipeline value	£649.6M
Completion model	Calibrated Random Forest (isotonic regression)
Timing model	Gradient Boosting (log-transformed outcome)
Noise injection	Gaussian on log scale, σ = model RMSE
Mean expected cashflow	£80.7M
90% confidence interval	£76.0M – £85.5M
Stress scenario (−10pp)	£69.8M
Recommended liquidity buffer	£10.9M

Table A4. Bootstrap simulation parameters and key outputs.

E. Note on the Analyst's Code Bug

The analyst's original code contains a one-line error in which the target variable y is assigned to the completion indicator (Complete) for model 1 and is never reassigned before model 2 is fitted:

```
y = data['Complete']          # set for model 1, never reassigned
time_to_completion_model = sma.OLS(y, sma.add_constant(X)).fit()  # predicts
boolean, not days
```

The OLS call therefore regresses a boolean completion flag (0 or 1) on the feature matrix, producing a misspecified linear probability model rather than a timing model. The RMSE of 11.41 days reported from this model is therefore not a valid timing baseline. The corrected baseline and OLS fitted on completed cases only with the same three features and no log transformation produces an RMSE of 13.52 days, which is used as the comparison figure throughout this report.